Taking a Trip – Using Planaria to Investigate Neuroplasticity and Learning Rates under the Influence of Psilocybin

By

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Special thanks to Professor Bart Ellenbroek for his supervision and guidance throughout, and to Kirsten Carter for her patience as I learned even the most basic lab skills. And of course, to Ella and my family, thank you for putting up with my stress and endless talk about mushrooms and worms.

Abstract

Objective: To investigate the impact of Psilocybin on learning rates in Planaria.

Background. Treatment-resistant depression remains a challenge, with therapies showing minimal efficacy. This highlights the need for novel interventions that enhance neuroplasticity and learning. Psilocybin, a serotonergic hallucinogen, shows promise in this regard but is challenging to study in humans and rodents due to ethical and practical constraints. Planarians, with their simple bilobed nervous system and diverse behavioural activity, provide an alternative model.

Methods. Firstly, psilocybin's optimal dose for locomotor activity changes was determined. Utilising planarians' innate negative phototaxis and responsiveness to appetitive cues, learning was assessed using a conditioned place preference (CPP) paradigm. Planarians were exposed daily to light, paired with sucrose for five consecutive days. Control groups included light-only and dark-only conditions for both psilocybin-treated and water groups. Statistical analysis employed ANOVA and t-tests to compare time and dose-dependent changes in behaviour.

Results. Psilocybin produced dose-dependent locomotor changes, with 5 μM identified as optimal. Psilocybin-treated planarians showed enhanced CPP learning compared to controls (time × group interaction: $F(1,12) = 5.86$, $p = 0.032$), with effects persisting five days post-treatment.

Conclusions. These findings suggest that psilocybin enhances learning in planarians, consistent with its proposed role in promoting neuroplasticity. This supports psilocybin's therapeutic potential.

Taking a Trip – Using Planaria to Investigate Neuroplasticity and Learning Rates under the Influence of Psilocybin

Psychedelic compounds such as psilocybin have re-emerged as a significant area of scientific research after decades of restricted study. Current research focuses on examining their therapeutic potential and the precise neurobiological mechanisms, with recent studies highlighting how these compounds produce both immediate and long-term changes in brain function and behaviour (Johnson & Griffiths, 2017).

Psilocybin: Pharmacology and Mechanisms

Psilocybin (4-phosphoryloxy-N, N-dimethyltryptamine) is a naturally occurring hallucinogenic prodrug that is dephosphorylated in the body to psilocin (4-hydroxy-N, N-dimethyltryptamine), the primary pharmacologically active compound (Passie et al., 2002). While classified as a hallucinogen, contemporary understanding recognises psilocybin as having complex neurobiological effects that extend beyond perceptual alterations, including influences on neuroplasticity and cognitive flexibility (Carhart-Harris & Friston., 2019). These neurobiological effects operate through multiple interconnected mechanisms across various levels of neural organisation. Psilocin acts primarily as an agonist at 5-HT2A receptors, which are widely distributed across cortical regions (Nichols., 2016). When activated by psilocin, these receptors initiate complex intracellular signalling cascades that influence neuronal excitability, gene expression, and structural plasticity. Psilocin increases expression of brain-derived neurotrophic factor (BDNF), a critical protein that promotes neuronal growth, synaptogenesis, and brain plasticity (Ly et al., 2018). This increase in BDNF occurs alongside elevated expression of immediate early genes such as c-fos and Arc, which are associated with neuronal activation and plasticity (Martin & Nichols., 2017). Through these mechanisms, psilocybin has been shown to enhance both the rate and extent of structural plasticity, promoting dendritic spine growth and synapse formation in cortical neurons in both human and animal studies (Ly et al., 2018).

Systems-Level Effects on Brain Networks

The functional consequences of these molecular cascades become apparent when examining connectivity within and between brain networks. Neuroimaging studies utilising fMRI have consistently demonstrated that psilocybin temporarily disrupts the activity and integrity of the default mode network (DMN), a collection of brain regions typically active during self-referential thinking (Carhart-Harris et al., 2012). This DMN disintegration occurs alongside increased global functional connectivity, creating a more entropic brain state with greater information exchange between normally segregated networks (Madsen et al., 2021; Carhart-Harris et al., 2014). This reduced hierarchical organisation allows novel connectivity patterns to emerge (Carhart-Harris & Friston, 2019). This neurobiological reorganisation may provide a mechanism that enables entrenched patterns of thought and behaviour to be reconsidered and potentially modified.

Clinical Applications and Therapeutic Potential

In humans, the effects of psilocybin manifest across multiple domains and time scales. Acute effects include perceptual alterations, emotional intensification, and cognitive changes that are dose-dependent and context-sensitive, with environmental setting and psychological preparation significantly influencing the experience (Griffiths et al., 2006; Griffiths et al., 2008). Psilocybin has also been shown to have persistent psychological effects that endure long after the compound has been metabolised and eliminated from the body. Follow-up studies of participants in psilocybin trials have found increases in personality openness which persist for at least 14 months after a single high-dose session (MacLean et al., 2011). This represents a degree of plasticity in personality traits that are typically considered stable in adulthood. Psilocybin has also demonstrated promising therapeutic effects in several clinical populations. In patients with treatment-resistant depression, two doses of psilocybin, alongside psychological support, produced substantial reductions in depression severity that were maintained for three months in many participants (Carhart-Harris et al., 2016). These persistent effects cannot be explained by the presence of the drug itself, as psilocin is eliminated from the body within 24 hours of administration (Dasgupta., 2019). Instead, they appear to result from neuroplastic changes initiated during the acute phase of administration that continue to develop and consolidate over subsequent weeks and months. Supporting this mechanism, preclinical studies have demonstrated that psilocybin can reduce neuronal loss and facilitate functional recovery in damaged neural tissue (Yu et al., 2024), suggesting its capacity to promote both neuroplasticity and neural repair.

The clinical significance of these findings becomes particularly apparent when considered against current mental health treatment challenges. Depression affects an estimated 280 million people worldwide (World Health Organisation, 2023), with approximately 30% of major depressive disorders proving treatment-resistant (Ionescu et al, 2015). Furthermore, remission rates following first- or second-course treatment reach only 37% and 31%, respectively (Rush et al., 2006). Against this backdrop, psilocybin's potential efficacy in treatment-resistant populations represents a possible paradigm shift in psychiatric care.

Preclinical Evidence in Mammalian Models

Recent preclinical evidence supports this potential. A study using a rat model of ischemic stroke found that psilocybin administration significantly reduced neuronal loss, improved locomotor function, and decreased neurological deficits, facilitating recovery through synaptic plasticity and neuroprotection (Yu et al., 2024). The capacity of psilocybin to enhance neurogenesis and modulate functional brain networks, such as the

DMN (Carhart-Harris et al., 2014), also suggests potential applications in neurodegenerative conditions such as Alzheimer's Disease (AD) (Zheng et al., 2024). Although direct studies on psilocybin in AD are limited, its ability to promote plasticity and reduce neuroinflammation could theoretically slow cognitive decline.

To establish causal relationships between psilocybin administration and observed effects, researchers often utilise mammalian models. Rodent studies have demonstrated that psilocybin and related psychedelics promote structural neuroplasticity, increasing dendritic spine density in cortical neurons within 24 hours of administration (Ly et al., 2018). This effect persisted for at least one week and was accompanied by enhanced neuritogenesis and spinogenesis in vitro. Further rodent studies have investigated the effects of psilocybin on learning and memory, with Catlow et al. (2013) finding that a moderate dose of psilocybin enhanced fear extinction learning in mice. Similarly, Hesselgrave et al. (2021) demonstrated that psilocybin produced antidepressant-like effects and enhanced synaptic plasticity in mice. These effects were accompanied by increases in dendritic spine formation in the mouse frontal cortex, establishing a direct link between structural plasticity and cognitive enhancement. Recent work by Shao et al. (2021) found that psilocybin increases the excitability of layer 5 pyramidal neurons and promotes dendritic spine growth in the mouse medial frontal cortex. These effects persisted for at least one week after a single dose. These findings collectively suggest that psilocybin's therapeutic potential stems from its ability to enhance neuroplasticity and facilitate adaptive behavioural changes across multiple domains of brain function.

While rodent models have provided valuable insights, they have notable limitations. Environmental factors, genetic variability, and complex social behaviours can significantly influence results (Hernandez & Blazer., 2006), making it challenging to isolate the specific effects of psychedelic compounds. To address these limitations, some researchers have begun exploring the effects of psychedelics in invertebrate models.

Planarians as an Alternative Model System

Planarians represent a particularly promising invertebrate model. These flatworms occupy a unique evolutionary position, possessing the simplest nervous system with bilateral symmetry and cephalisation, making them valuable for studying fundamental neural mechanisms (Quiroga et al, 2015). The planarian model offers several distinct advantages over both mammalian and other invertebrate models for investigating the persistent effects of psilocybin. Unlike mammalian models, planarians permit direct observation of behavioural changes in living organisms with minimal ethical concerns. Their simple nervous system facilitates a more straightforward interpretation of fundamental mechanisms that might be obscured by greater complexity in vertebrate brains. Compared to other invertebrate models such as Drosophila or C. elegans, planarians possess a more centralised nervous system with greater relevance to vertebrate brain organisation, while still allowing experimental control (Brown & Pearson., 2017). Their capacity for associative learning allows for assessment of cognitive effects before, during, and after drug exposure. Furthermore, planarians adhere to the ‘3Rs’ principles (Replacement, Reduction, and Refinement) (Russell., 2005). As planarians are not currently considered capable of experiencing suffering, their use satisfies the Replacement principle by providing an alternative to sentient animals in psychedelic research.

The planarian nervous system consists of approximately 20,000 neurons organised into a primitive brain with distinct functional regions, including structures analogous to the vertebrate dopaminergic and serotonergic systems (Buttarelli et al., 2008). Planarians possess multiple serotonin receptor subtypes that share significant homology with mammalian receptors, including those that mediate psychedelic effects (Farrell et al, 2008). Importantly, planarians express receptors analogous to the 5-HT2A receptor which is the primary target of psilocin in mammals (Madsen et al., 2019). This allows for meaningful investigation of drug effects that may translate to more complex organisms. The use of planarians in pharmacological research is relatively recent. Studies demonstrate their sensitivity to drugs affecting monoaminergic systems, with planarians showing characteristic behavioural responses to serotonergic compounds, including changes in locomotor activity and specific movement patterns (Buttarelli et al., 2000). These responses are often dose-dependent and can be blocked by appropriate receptor antagonists (Palladini et al., 1996), suggesting pharmacological specificity similar to that observed in vertebrates. While relatively few studies have specifically examined psychedelic compounds in planarians, related work provides support for their utility.

Despite their relative simplicity, planarians display a remarkable ability to learn behaviour. Classical conditioning studies have demonstrated that planarians can be trained to respond to specific stimuli, including light and dark (McConnell, 1966; Jacobson, 1967). This learning ability provides researchers with a unique opportunity to investigate basic neuroplasticity and cognitive processes at a foundational level. Furthermore, planarians exhibit negative phototaxis. Their light-aversion mechanism represents a critical survival strategy characterised by directional movement away from light sources and involves neural processing in their primitive brain. Experimental studies have shown that planarians can adaptively modify their light-avoidance behaviour through repeated exposure. Zewde et al. (2018) investigated planarian light aversion using an experimental design with controlled light intensity gradients and high-resolution tracking systems. The researchers exposed planarians to varying light conditions and measured their movement patterns, response times, and behavioural modifications. They found that planarians can adaptively modify their light-avoidance behaviour through repeated exposure. Planarians also demonstrate sensitivity to appetitive stimuli, with sucrose serving as an effective positive reinforcer. Research has shown that planarians exhibit preference behaviour for sucrose solutions and can learn to associate environmental cues with sucrose availability (Buttarelli et al., 2000). This capacity for reward-based learning makes it possible to investigate whether psilocybin enhances the formation of positive associations, providing a direct parallel to therapeutic contexts where patients learn new adaptive behavioural patterns.

Study Rationale and Hypotheses

Despite the growing body of evidence regarding psilocybin's acute and persistent effects in humans and rodents, significant gaps remain in our understanding of these effects and thus their potential applications in clinical populations. The current study takes a step forward in this field by examining learning following psilocybin exposure in planarians, leveraging this model's advantages to isolate and characterise basic mechanisms underlying psychedelic-induced neuroplasticity. By establishing the time course of recovery from acute locomotor effects and subsequently assessing learning capacity across multiple timepoints, this research will help distinguish between direct pharmacological actions and the neuroplastic changes that persist beyond drug elimination. This experimental approach parallels the clinical literature in humans, where therapeutic benefits often emerge and strengthen after acute drug effects have completely subsided.

Based on the evidence for psilocybin's neuroplasticity-enhancing effects across mammalian species and the evolutionary conservation of serotonergic signalling mechanisms, it is hypothesised that psilocybin will enhance learning and behavioural plasticity in planarians. Specifically, we predict that planarians exposed to psilocybin will demonstrate a significantly greater capacity for associative learning than control subjects. Further, we expect these enhanced learning effects to persist beyond the drug's acute pharmacological action, consistent with underlying neuroplastic changes rather than direct drug effects. This study aims to establish optimal dosing parameters by determining the dose-response relationship for psilocybin in planarian locomotor behaviour to identify a concentration that produces measurable behavioural effects without toxicity. We further aim to investigate whether psilocybin pre-treatment enhances the acquisition of conditioned place preference behaviour in planarians using a light-sucrose association paradigm. Through this investigation, we hope to contribute insights into the neurobiological mechanisms that may explain psilocybin's therapeutic potential, from the simple neural circuits of planarians to the complex therapeutic mechanisms now emerging in human medicine.

Method

Study design

This controlled experimental study employed a between-subjects design to examine psilocybin's effects on associative learning in planarians. The primary comparison involved psilocybin-treated versus control groups undergoing light-sucrose conditioning, with additional light-only and dark exposure control conditions to isolate specific learning mechanisms. Following a single psilocybin exposure, planarians underwent a 5-day conditioning protocol, with the primary outcome being the change in light preference ratio from baseline to post-conditioning assessment.

Colony

Freshwater Planarians were used for this study. As New Zealand regulations prohibit the import of Planarian species, planaria were collected from a local stream in Wellington. Initial observations suggest the specimens may be *Neppia*, a genus native to New Zealand. Species-level identification is ongoing, but the exact species has not been confirmed. The collected specimens appeared relatively homogenous in size and colouration (light brown), though minor individual variation was observed. Planarians were kept under laboratory conditions in their native spring water at 20°C. Planarians were fed raw beef liver once weekly, except during a 4-day food deprivation period prior to experimental procedures to standardise feeding motivation. Ethical approval was not required for this research.

Materials and Apparatus

Housing and Testing Environment

For standard housing, planarians were kept individually in 5 cm Petri dishes containing 7 mL of spring water. During testing, identical Petri dishes were modified by attaching double-layered tin foil to half of the dish exterior to produce a distinct light/dark environment for use in the conditioned place preference (CPP) paradigm.

Pharmacological Agents

Psilocybin, obtained from BDG Synthesis, was dissolved in deionised water to create a 1000 µM stock solution. This solution was diluted with filtered spring water to produce the required concentrations for behavioural testing. Control solutions consisted of spring water without psilocybin.

Behavioural Tracking System

Locomotor and place preference behaviour was recorded using EthoVision XT 15.0 (Noldus Information Technology) connected to a digital camera mounted 30 centimetres above the test arena. An LED backlight panel provided uniform illumination.

Procedure

Dose-Response Assessment

A dose-response study was conducted to identify the optimal psilocybin concentration for use in the main experiment. Thirty planarians were randomly assigned to six treatment groups ($n = 5$ per group) receiving 0, 0.5, 5, 10, 20, or 50 µM psilocybin.

Each planarian was placed individually in a 5 cm Petri dish containing 7 mL of the respective test solution. Behaviour was recorded continuously for four hours. At hourly intervals, planarians were gently startled using a Pasteur pipette to assess recovery and activity responses. Locomotor parameters were quantified using EthoVision software. Based on the consistency and clarity of behavioural responses, 5 µM psilocybin was selected as the optimal dose for the main experiment.

Main Experiment

Group Assignment. Forty-eight planarians were randomly assigned to one of six experimental groups (n = 8 per group). The experimental manipulations for each group can be seen in Figure 1.

Figure 1

Planned Experimental Timeline and Procedures

Group	Day 0 Drug Administration	Day 1 Baseline	Day 2 – 6 Conditioning	Day 7 Post-Test
PSILO-T N=8	5µM Psilocybin 4 hours in solution + 24 hours recovery	Light/Dark Chamber 2 min acclimatation, 10-minute recording	**Conditioning** LED light + 10% sucrose 10 min/day 7 mL solution	Light/Dark Chamber 2 min acclimatation, 10-minute recording
PSILO-L N=8	5µM Psilocybin 4 hours in solution + 24 hours recovery	Same as above	**Light only** LED light + spring water 10 min/day No sucrose	Same as above
PSILO-D N=8	5µM Psilocybin 4 hours in solution + 24 hours recovery	Same as above	**Dark Exposure** Complete darkness Spring water only 10 min/day	Same as above
CTRL-T N=8	Spring Water 4 hours in petri dish + 24 hours recovery	Light/Dark Chamber 2 min acclimatation, 10-minute recording	**Conditioning** Same as PSILO-T	Light/Dark Chamber 2 min acclimatation, 10-minute recording
CTRL-L N=8	Spring Water 4 hours in petri dish + 24 hours recovery	Same as above	**Light only** Same as PSILO-L	Same as above
CTRL-D N=8	Spring Water 4 hours in petri dish + 24 hours recovery	Same as above	**Dark Exposure** Same as PSILO-D	Same as above

Drug Administration. Drug exposure occurred in 5 cm Petri dishes containing 7 mL of solution. Planarians in psilocybin groups received 5 µM psilocybin, while controls received equivalent volumes of spring water. All planarians were exposed to their respective solutions for 4 hours and then returned to their home containers for 24 hours to allow acute drug effects to dissipate.

Baseline Preference Assessment. Twenty-four hours after drug administration, each planarian was individually placed in the centre of a 5 cm Petri dish. Following a two-minute acclimation period, double-layered tin foil was attached to half of the dish, creating a split light/dark dish. Behaviour was then recorded for 10 minutes. Due to software limitations, time spent in the light compartment was measured manually, with time in the dark compartment inferred by subtracting this value from the total test duration. The primary behavioural metric was the proportion of time spent in the light versus dark compartments.

Conditioning Phase. The experimental timeline and procedures for all groups are summarised in Figure 1. The conditioning phase commenced 24 hours after baseline testing and continued for five consecutive days. Each group followed one of three distinct conditioning protocols as detailed below:

Treatment groups (PSILO-T and CTRL-T): Planarians were placed in petri dishes on the LED light panel and exposed to 7 mL of 10% sucrose solution for 10 minutes daily. This protocol aimed to establish a positive association between light exposure and appetitive reward.

Light-only groups (PSILO-L and CTRL-L): Planarians were exposed to spring water only under identical lighting conditions for 10 minutes daily. This protocol controlled for light exposure effects without appetitive reinforcement.

Dark control groups (PSILO-D and CTRL-D): Planarians were exposed to spring water in complete darkness for 10 minutes daily. This protocol served as a baseline condition with minimal environmental stimulation.

After each conditioning session, all planarians were returned to their individual home containers until the following day's session.

Post-Conditioning Assessment and Extinction. The day following the final conditioning session, post-conditioning testing was conducted using the same procedure as the baseline assessment. Each planarian was placed in a light dish for a two-minute acclimation period, followed by a ten-minute recording period in a split dish.

Data Analysis

Dose Response Study

Behavioural Metrics. The primary outcome measure was locomotor activity of the Planarians. This was recorded using EthoVision Software and using millimetres (mm) as the unit of measurement.

Statistical Analysis. Based on previous work in the lab, it was hypothesised that 5 μM would be the preferred dose, showing a more consistent response to startling procedures. Group differences in locomotion distance were analysed using a one-way analysis of variance (ANOVA), with treatment group as the independent variable. Following a significant ANOVA, post hoc pairwise comparisons were conducted between each treatment group and the control group. Tukey's Honest Significant Difference (HSD) test was used to control for Type I error across multiple comparisons. All analyses were performed in Python. Significance is reported at an alpha level of 0.05. All data analysis was run twice to minimise potential for error.

Main Experiment

Behavioural Metrics. The primary outcome measure was the preference ratio, calculated as the time spent in the light compartment divided by total time.

Statistical Analysis. Based on the predicted enhancement of neuroplasticity and associative learning capacity, we hypothesised that planarians in the PSILO-T group would demonstrate significantly higher preference ratios (time in light compartment/total time) compared to the CTRL-T group following conditioning (H1: μPSILO-T > μCTRL-T, $p < 0.05$). It was further predicted that a significant time × group interaction for treatment (T) groups would be seen, with the psilocybin-treated group (PSILO-T) expected to show a greater increase in light preference from baseline to post-conditioning compared to the control group (CTRL-T). Secondary hypotheses included that psilocybin groups exposed to light alone (PSILO-L) would show enhanced behavioural flexibility compared to corresponding controls (CTRL-L).

Baseline preference scores were analysed using one-sample t-tests against the chance level (preference ratio = 0.5) to assess innate preferences. Between-group differences were analysed using two-way repeated measures ANOVA, with Group (PSILO vs CTRL) as a between-subjects factor and Phase (Baseline vs Post-conditioning) as a within-subjects factor. Where appropriate, significant interactions were followed by Bonferroni-corrected post hoc tests. Effect sizes were reported to assess the practical significance of effects. Statistical significance was set at $p < 0.05$. All analyses were conducted using Python and Jamovi. All data analysis was run twice to minimise potential for error.

Quality Control Measures. Quality control measures were implemented throughout the study. Planarians showing abnormal mobility during the baseline assessment were excluded from further testing as part of the study design; however, no planarians were excluded on this basis. Environmental conditions were standardised across all testing days to minimise extraneous variables. All testing was conducted in a private room within the TTR L0 PC2 facility, allowing for consistent control of temperature (maintained at 20°C), lighting (using the LED backlight panel), and timing (all sessions conducted between 10:00-16:00). Four subjects died during the experimental period and were excluded from analysis: two subjects from CTRL-D (subjects 7 and 8), one subject from CTRL-T (subject 7), and one subject from PSILO-L (subject 1). Mortality appeared to occur randomly across groups with no concentration in psilocybin-treated conditions, suggesting deaths were unrelated to drug treatment. Final group sizes ranged from 6 to 8 subjects, maintaining adequate statistical power for the planned analyses.

Results

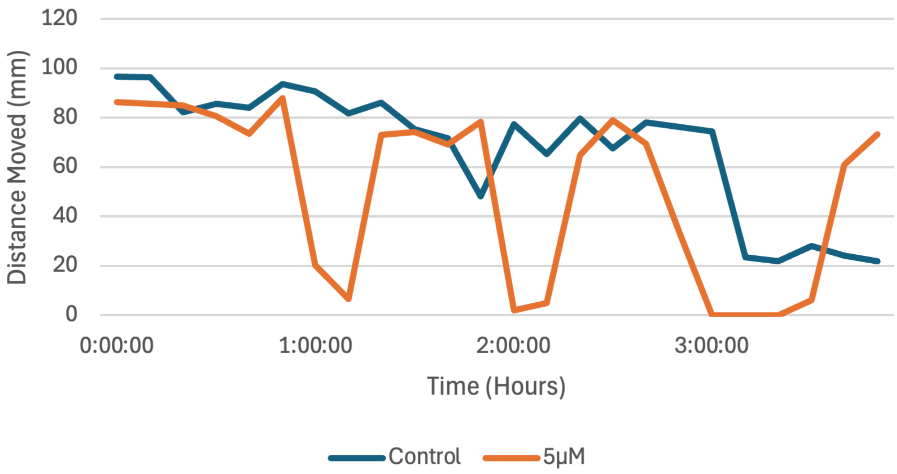
Dose-Response Effects of Psilocybin on Planaria Locomotion

Planarian locomotor activity was recorded over four hours following exposure to six psilocybin concentrations ($n = 30$; 5 per group). A one-way ANOVA revealed a significant main effect of group on the outcome variable, $F(5, 138) = 6.04$, $p < .001$. However, post-hoc Tukey comparisons between the Control condition and each treatment group showed no statistically significant differences (all $p > .05$). The most considerable difference was observed between the 0.5 μM group and Control (mean difference = 25.85, 95% CI [-1.41, 53.12], $p = .074$), which approached significance but did not survive correction. All other comparisons with Control were non-significant (5: $p = .45$; 10: $p = .98$; 20: $p = .36$; 50: $p = .95$).

Figure 2 illustrates the temporal pattern of locomotor changes across the four-hour observation period. The biphasic response pattern observed at 5 μM concentration shows an initial decrease in activity followed by recovery, with consistent responsiveness to startling procedures throughout the observation period. Saline exposure did not produce any effects on locomotor behaviour. The figures for the other doses are in Appendix A.

Figure 2

Graph Showing Movement of Planaria Over Time for 5 µM and Control Groups



The significant overall effect without post hoc differences likely reflects the conservative nature of Tukey corrections with small sample sizes (n=5 per group). The observational consistency of 5 µM responses supported its selection despite statistical ambiguity. This choice proved successful in demonstrating clear learning effects in the main experiment, validating the methodological approach

Main Experiment

Baseline

One-sample t-tests were conducted to compare the proportion of time spent in the light compartment against chance levels (0.5) for each experimental group. All groups demonstrated a preference for the dark compartment, with mean values ranging from 0.274 to 0.392 (see Table 1). Four groups showed statistically significant light avoidance behaviour: CTRL D ($p < 0.001$), CTRL T ($p < 0.001$), PSILO L ($p = 0.002$), and PSILO T ($p = 0.005$). Two groups approached but did not reach statistical significance: CTRL L ($p = 0.054$) and PSILO D ($p = 0.053$). All effect sizes were large ($|d| > 0.8$), indicating substantial behavioural differences from chance performance across all experimental conditions

Table 1

Light Preference Ratio Compared to Chance Level (0.5)

Group	n	Mean ± SD	p-value	Cohen's d
CTRL – D	8	0.350 ± 0.072	<0.001	-2.08
CTRL – L	8	0.361 ± 0.170	0.054	-0.82
CTRL – T	8	0.370 ± 0.067	<0.001	-1.94
PSILO – D	8	0.392 ± 0.131	0.053	-0.82
PSILO – L	8	0.274 ± 0.137	0.002	-1.65
PSILO – T	8	0.340 ± 0.112	0.005	-1.42

Conditioning Effects: Primary Analysis

Following the five-day conditioning protocol, distinct patterns of behavioural change emerged across the six experimental groups (see Table 2 and Figure 3). After exclusions for mortality (see Methods), the movement of 44 planarians was assessed following the five-day conditioning protocol. Distinct patterns of behavioural change emerged across the six experimental groups (see Table 2 and Figure 3).

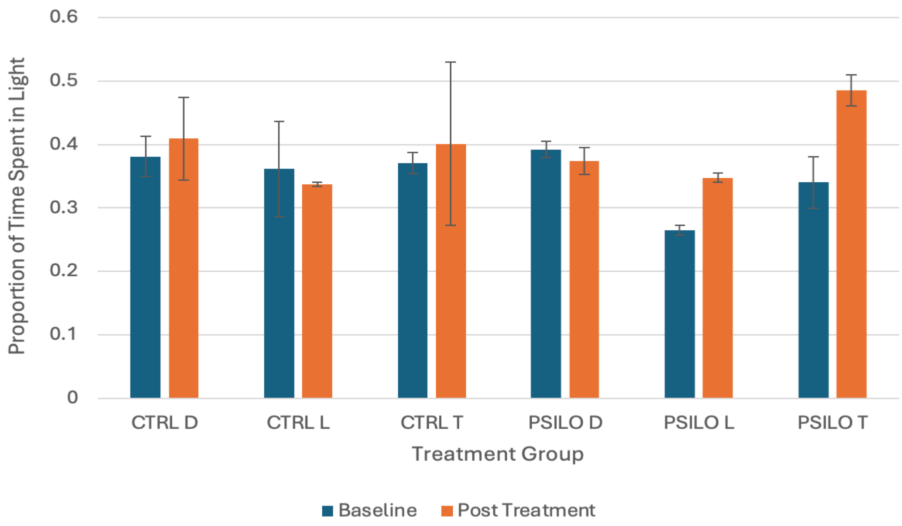
Table 2

Mean and SD of Movement across Time Phases

Group	N	Baseline M (SD)	Post-Treatment M (SD)	% Change	Effect Magnitude
CTRL-D	6	0.381 (0.052)	0.409 (0.125)	+7.40	Small
CTRL-L	8	0.361 (0.169)	0.337 (0.113)	-6.75	Small
CTRL-T	7	0.371 (0.072)	0.400 (0.170)	+8.21	Small
PSILO-D	8	0.392 (0.131)	0.374 (0.108)	-4.59	Small
PSILO-L	7	0.265 (0.145)	0.348 (0.068)	+31.27	Large
PSILO-T	8	0.340 (0.112)	0.485 (0.078)	+42.65	Large

Figure 3

Percentage of Time Spent in Light Compartment in Baseline and Post-Treatment Trials



Control groups showed minimal behavioural changes regardless of conditioning protocol: CTRL-D exhibited a slight increase in light preference (+7.40%), CTRL-L showed a slight decrease (-6.75%), and CTRL-T demonstrated a modest increase (+8.21%). Using one-sample t-tests and Cohen's d, it was found that all control group changes were classified as having small effect magnitudes, suggesting a limited impact of conditioning protocols in the absence of psilocybin.

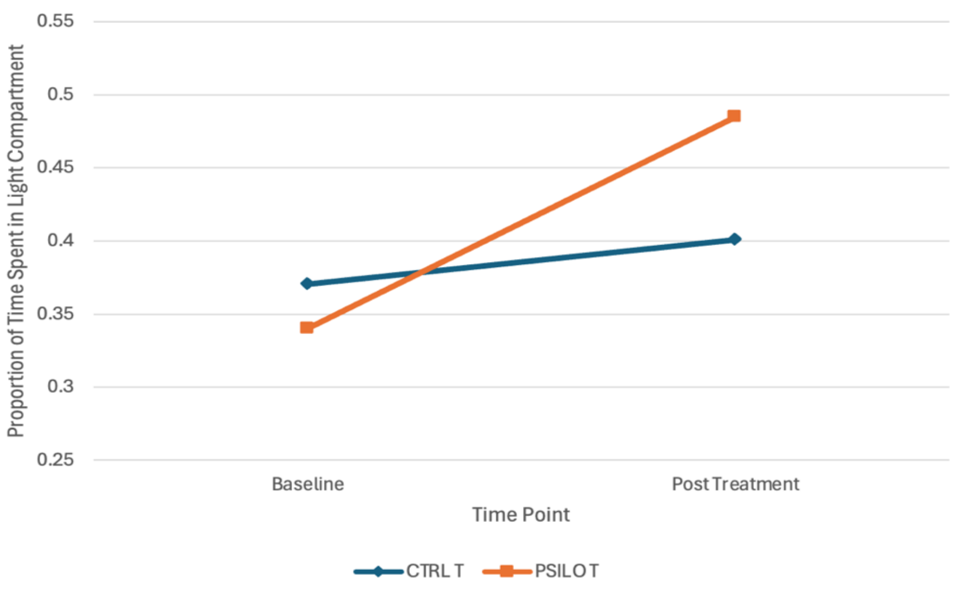
In contrast, psilocybin-treated groups showed more variable and pronounced responses. PSILO-D exhibited a slight decrease in light preference (-4.59%), indicating that psilocybin exposure alone without appetitive conditioning did not enhance light preference. However, both psilocybin groups exposed to light during conditioning showed substantial increases: PSILO-L increased by 31.27% and PSILO-T increased by 42.65%, both classified as large effect magnitudes. This pattern suggests that psilocybin enhanced behavioural plasticity specifically in response to light-associated experiences, with the strongest effects occurring when light exposure was paired with appetitive reinforcement (sucrose).

Primary Statistical Analysis: CTRL-T versus PSILO-T

A repeated-measures ANOVA comparing the two primary groups of interest (CTRL-T vs PSILO-T) revealed a significant effect of psilocybin. There was a significant main effect of time ($F(1,12) = 11.08$, $p = 0.006$), indicating overall changes in light preference from baseline to post-conditioning across all worms. Critically, there was a significant time × group interaction ($F(1,12) = 5.86$, $p = 0.032$), demonstrating that psilocybin and control groups responded differently to the conditioning protocol (Figure 4).

Figure 4

Graph Showing Temporal Change in Light Preference for PSILO-T and CTRL-T Groups



Examination of the means revealed that the CTRL-T group showed a modest increase in light preference from baseline (M = 37.1%) to post-conditioning (M = 40.0%), representing a small effect magnitude (as per Table 2). In contrast, the PSILO-T group demonstrated a substantially larger increase from baseline (M = 34.0%) to post-conditioning (M = 48.5%), representing a 42.65% relative increase. This pattern confirms that psilocybin significantly enhanced the acquisition of conditioned light preference when paired with sucrose reward.

Discussion

This study investigated whether psilocybin enhances associative learning capacity in planarians. The results demonstrate that psilocybin-treated planarians showed significantly greater learning capacity compared to controls, providing evidence that psychedelic-induced neuroplasticity operates through evolutionarily conserved mechanisms. A significant time × group interaction ($F(1,12) = 5.86$, $p = 0.032$) was observed, demonstrating that psilocybin altered the animals' ability to modify innate behavioural preferences through experience during appetitive conditioning. These effects persisted five days post-administration, well beyond the timeframe of acute drug action, suggesting lasting neuroadaptive changes rather than transient pharmacological effects. The following sections explore these findings in relation to appetitive learning mechanisms, temporal dynamics of neuroplasticity, neurobiological mechanisms, and broader implications for psychedelic research.

Psilocybin Enhancement of Appetitive Learning

While control planarians showed minimal change in their natural light avoidance after light-sucrose pairing, psilocybin-treated planarians displayed notable behavioural plasticity. Baseline testing confirmed consistent light avoidance across all groups, establishing that subsequent modifications represent alterations of innate phototactic responses rather than baseline differences. Psilocybin-treated planarians exhibited enhanced learning in both the light-reward condition (PSILO-T: +42.65%) and the light-only condition (PSILO-L: +31.27%), with the most pronounced effect when light exposure was paired with appetitive reinforcement.

The PSILO-T group's 42.65% increase represents the core finding, where psilocybin increases appetitive associative learning when reward contingencies are present. This demonstrates its capacity to facilitate the formation of new stimulus-reward associations that reshape innate behaviours. The PSILO-L group also showed enhancement (+31.27%) despite receiving no appetitive reinforcement, suggesting psilocybin's learning effects extend beyond reward-based conditioning to include habituation or extinction-like processes. However, the additional 11.38 percentage point benefit observed with reward pairing confirms that psilocybin potentiates appetitive learning mechanisms. This broad enhancement of learning capacity, rather than simple reduction of aversive responses, represents a shift in behavioural flexibility that surpasses specific conditioning paradigms.

These findings parallel mammalian psychedelic research, demonstrating enhanced learning capacity across multiple domains. Catlow et al. (2013) showed that psilocybin enhanced fear extinction learning in mice, enabling animals to modify conditioned fear responses through experience. The analogous processes observed across phyla suggest conserved mechanisms underlying psychedelic-induced learning enhancement.

Temporal Dynamics

The five-day persistence of enhanced learning capacity distinguishes neuroplastic adaptations from direct pharmacological effects. Psilocin is eliminated from biological systems within 24 hours of administration (Dasgupta, 2019), yet the behavioural changes observed in our study emerged during testing conducted five days post-exposure. This indicates that the enhanced learning capacity reflects lasting changes in neural function rather than the continued presence of the drug. This parallels the temporal dynamics observed in mammalian studies. Hesselgrave et al. (2021) found that psilocybin produced antidepressant-like effects and enhanced synaptic plasticity that persisted well beyond the acute presence of the drug in mice. Shao et al. (2021) demonstrated that structural neuroplasticity changes in the mouse cortex, including increased dendritic spine density and pyramidal neuron excitability, continued for at least one week following a single psilocybin dose. The convergence of these findings across species suggests that psychedelics initiate neuroplastic processes during acute administration that continue to develop and consolidate over subsequent days and weeks.

In humans, this pattern manifests as therapeutic benefits that often strengthen after acute drug effects have subsided. Follow-up studies have found that increases in personality openness persist for at least 14 months after a single high-dose psilocybin session (MacLean et al., 2011), while patients with treatment-resistant depression showed substantial symptom reductions maintained for three months following treatment (Carhart-Harris et al., 2016). The persistence of effects in our planarian model is notable given the planarian nervous system's relative simplicity compared to mammalian brains. This suggests that the temporal dynamics of psychedelic-induced neuroplasticity do not depend on complex cortical networks but operate through more fundamental mechanisms.

Neurobiological Mechanisms

The behavioural changes we observed likely occur through mechanisms that share features with those characterised in mammalian systems. In mammals, psilocybin's effects involve cascades initiated through 5-HT2A receptor activation, leading to increased BDNF expression, immediate early gene activation, and structural synaptic modifications that promote dendritic spine growth and synaptogenesis (Ly et al., 2018). The presence of serotonin receptor subtypes in planarians that share homology with mammalian receptors, including those analogous to the 5-HT2A receptor (Farrell et al., 2008), provides a mechanistic foundation for our findings.

However, the gap between planarian and mammalian nervous systems requires careful interpretation. Mammalian psychedelic effects involve complex interactions between cortical networks, neurotransmitter systems, and cognitive processes (Hatzipantelis & Olson, 2023). Neuroimaging studies have shown that psilocybin temporarily disrupts the DMN while increasing global functional connectivity, creating a more integrated pattern of brain activity (Carhart-Harris et al., 2012; Madsen et al., 2021). The behavioural flexibility we observed in planarians likely reflects changes in simpler neural circuits within their bilobed nervous system rather than network-level reorganisation. Despite this difference in complexity, the behavioural outcomes appear similar across species, suggesting that neuroplasticity enhancement may be a fundamental property of serotonergic psychedelics that operates at multiple levels of neural organisation and across phylogenetically diverse species. In mammalian systems, these effects operate at multiple levels of neural organisation. Receptor activation initiates intracellular signalling cascades at the cellular level, leading to modifications of synaptic connections and neural excitability at the circuit level, and reorganisation of network connectivity patterns at the systems level. The planarian model, with its simpler nervous system, may help isolate foundational mechanisms by reducing the confounding complexity inherent in mammalian cortical networks.

Methodological Considerations and Study Limitations

Several methodological strengths increase confidence in our findings. The dose-response study identified 5 μM psilocybin as producing optimal behavioural effects without apparent toxicity. The 24-hour washout period ensured observed changes reflected neuroplastic adaptations rather than acute drug effects. The experimental design incorporated appropriate controls (drug-alone, light-alone, and combined conditions) that ruled out alternative explanations, while baseline measurements confirmed normal phototactic behaviour.

However, several important limitations suggest clear directions for future research. The most significant concern is species identification. While initial observations suggest the specimens may be *Neppia*, definitive taxonomic identification is incomplete. This limits generalisability, as different planarian species may exhibit varying sensitivity to serotonergic compounds. However, this limitation is unlikely to impact our core conclusions, as fundamental serotonergic signalling mechanisms are highly conserved across planarian species. Future work should prioritise taxonomic identification and, ideally, compare responses across multiple planarian species to establish the generalisability of these effects.

A second concern relates to the manual scoring of behavioural preferences. Although the experimenter was blind to the treatment condition during scoring to minimise this risk, the use of a more advanced automated system would eliminate this potential for bias.

Additionally, our preliminary dose-response experiment did not yield statistically significant differences between individual concentrations in post hoc comparisons, despite identifying 5 μM as optimal based on observational criteria. While this choice successfully demonstrated behavioural effects in the main experiment, the lack of clear dose-dependent statistical significance in individual comparisons reduces confidence in identifying the precise optimal concentration. Future dose-response studies with larger sample sizes, finer concentration gradients, and longer observation periods would strengthen characterisation of the dose-response relationship and provide greater statistical power to identify concentration-dependent effects. Mapping of these parameters would also allow comparison across studies and facilitate translation to other model systems.

Our experimental timeline, while sufficient to demonstrate persistence of effects beyond acute drug presence, did not capture the full temporal dynamics of plasticity enhancement. Extending the observational timeline beyond five days would clarify the full duration of enhanced learning capacity and whether effects eventually decay or remain stable. Implementing repeated dosing paradigms could reveal whether multiple psilocybin exposures produce cumulative enhancement of neuroplasticity or whether ceiling effects limit additional benefit.

Our behavioural approach could not directly examine the cellular and molecular changes underlying enhanced plasticity. The specific molecular cascades downstream of receptor activation in planarian neurons remain unknown. Without receptor-specific antagonists or molecular assays, determining whether homologous 5-HT2A pathways mediate the observed effects remains speculative. While mammalian literature suggests involvement of 5-HT2A receptor pathways, BDNF upregulation, and structural synaptic modifications, whether analogous mechanisms mediate the observed effects in planarians cannot be determined from our current data. Additionally, while our findings demonstrate enhanced neuroplasticity, we cannot make specific claims about neurogenesis, as our behavioural methodology does not directly assess this. Combining behavioural analysis with molecular and cellular techniques would bridge this gap. Pharmacological tools (5-HT2A antagonists), molecular approaches (RNA sequencing, immunohistochemistry for BDNF-like proteins), and advanced imaging could confirm receptor specificity, identify gene expression changes, and visualise structural modifications in planarian neural networks.

Furthermore, future extinction learning studies could examine whether psilocybin enhances the unlearning of maladaptive associations, directly addressing mechanisms relevant to PTSD and phobias. Training planarians to avoid neutral stimuli through aversive conditioning, then testing psilocybin's effects on overcoming these responses when threats are removed, would distinguish whether psilocybin facilitates new learning or primarily reduces existing aversions. The development of planarian models for investigating different learning types could reveal whether psilocybin's effects extend beyond associative learning to more sophisticated cognitive processes, helping determine the breadth of cognitive domains affected by psychedelic treatment.

Implications for Understanding Psychedelic Therapeutics

The findings from this study contribute to the understanding of psychedelics as neuroplasticity enhancers rather than purely perceptual-altering compounds. Contemporary research increasingly recognises that therapeutic benefits stem from effects on neural structure and function that persist beyond acute phenomenological experiences (Johnson & Griffiths, 2017). Our demonstration that psilocybin can enhance behavioural plasticity in organisms with simple nervous systems supports this framework by showing that plasticity enhancement operates through neurobiological mechanisms conserved across evolutionary lineages.

The clinical relevance of these findings becomes apparent when considered against the backdrop of current mental health treatment challenges. The suggestion that psychedelics enhance the brain's capacity for adaptive change offers a novel therapeutic approach that could address the underlying cognitive and behavioural rigidity characterising many mental health conditions. If psilocybin's therapeutic benefits stem from enhancements to neural adaptability, then the conservation of these effects across evolutionary lineages suggests that similar mechanisms might be therapeutically accessible across diverse clinical populations. While neurobiological differences require caution in direct clinical translation, these findings inform mechanistic understanding. The temporal pattern observed mirrors clinical observations that therapeutic benefits strengthen after acute effects subside. This persistent increase in cognitive flexibility highlights psilocybin's potential to treat Depression and anxiety disorders which are characterised by rigid thought and behaviour patterns resistant to modification (Kashdan & Rottenberg., 2010). If psilocybin enhances adaptive capacity through mechanisms conserved from planarians to humans, therapeutic benefits may arise from fundamental biological properties rather than complex cognitive processes, potentially explaining efficacy across diverse patient populations. If confirmed through additional research, this perspective could support the development of psychedelic-assisted interventions specifically designed to enhance neuroplasticity windows for therapeutic benefit.

Conclusions

This study demonstrated that psilocybin enhances associative learning in planarians, providing evidence that psychedelic-induced neuroplasticity operates through evolutionarily conserved mechanisms present in even the simplest bilateral nervous systems. The learning improvements observed, combined with robust statistical evidence and persistence beyond acute drug effects, establish that therapeutic benefits are not dependent on complex mammalian-specific cognitive processes but rather operate through simple neural pathways.

These findings have implications for understanding psychedelic-assisted therapy, suggesting that the therapeutic benefits observed in human clinical trials result from enhancements to the brain's capacity for adaptive change. The conservation of these mechanisms from planarians to humans provides evidence that psilocybin taps into universal biological processes for promoting behavioural flexibility and learning. This represents an important step toward understanding how psychedelic compounds enhance the brain's capacity for adaptive change across phylogenetically diverse species, contributing to our broader understanding of the biological foundations of learning, memory, and behavioural flexibility that has the potential to inform therapeutic applications in human populations.

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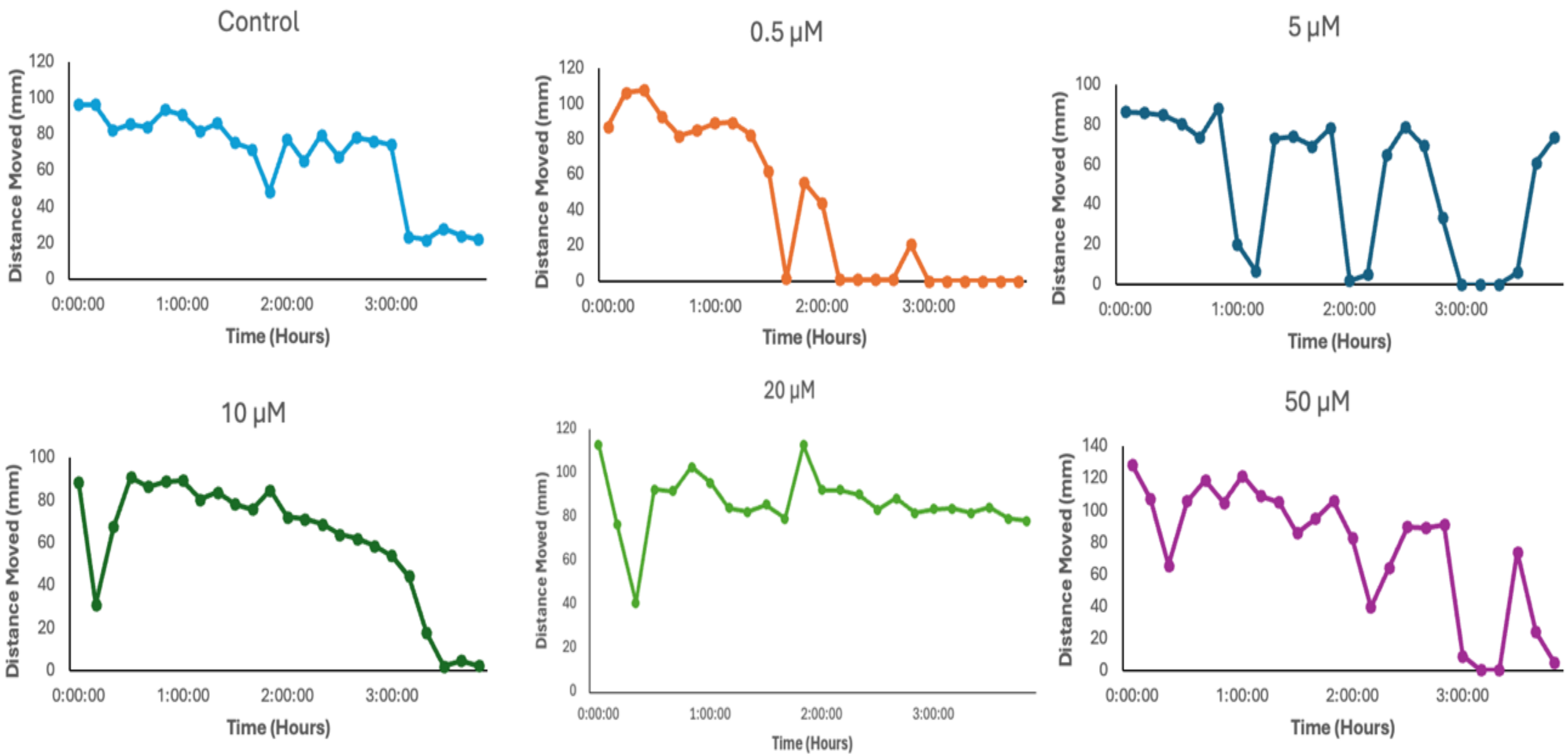
Appendix



Locomotor activity over time across Psilocybin concentrations.

Distance moved (in millimetres) was recorded over a 4-hour period, with time shown in hours. Each subplot represents one of six treatment groups: Control (0 µM), 0.5 µM, 5 µM, 10 µM, 20 µM, and 50 µM Psilocybin. The data illustrate dose-dependent effects on movement patterns, with notable decreases in locomotor activity at specific concentrations.